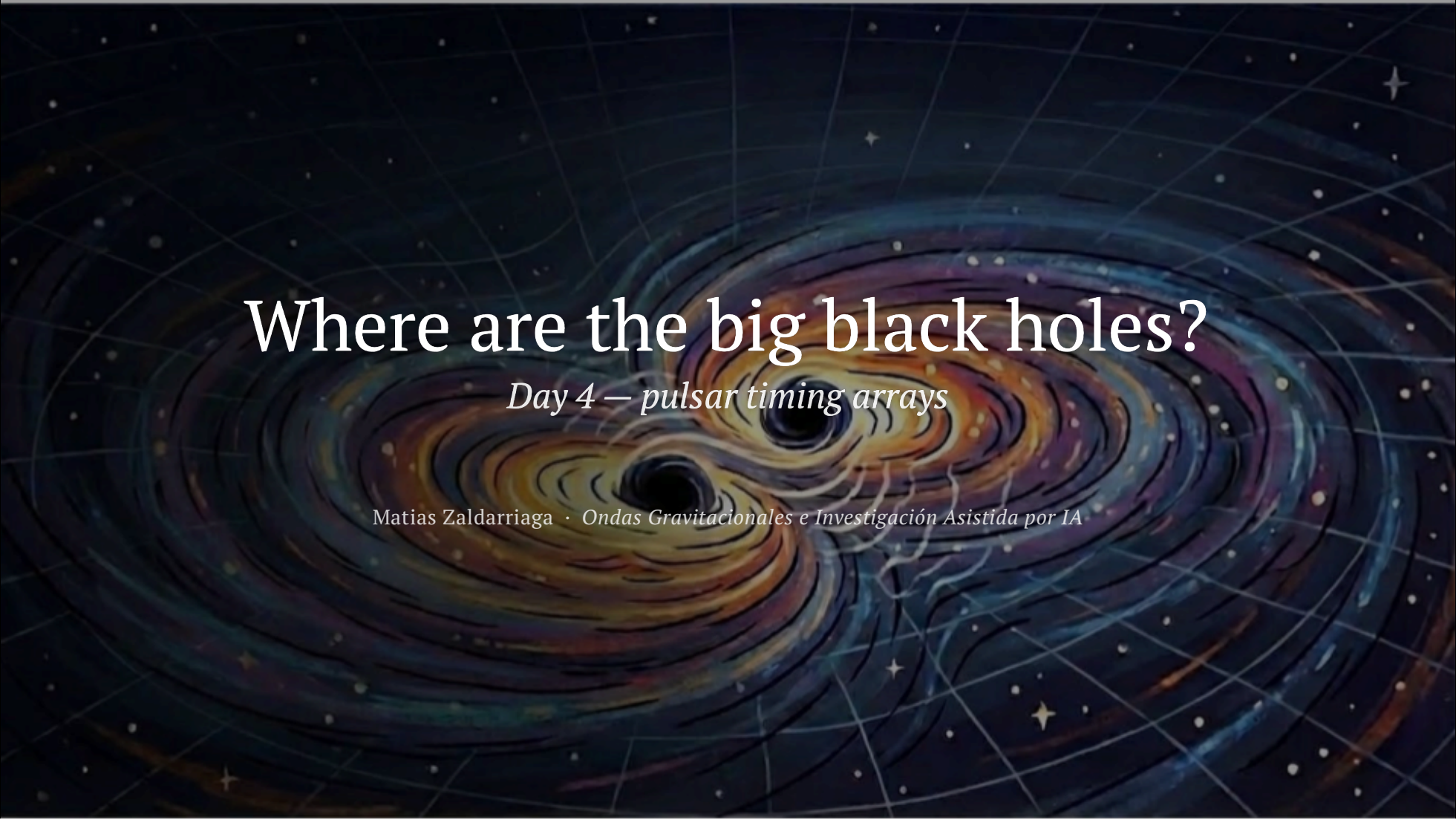




Where are the big black holes?

Day 4 — pulsar timing arrays

Matias Zaldarriaga · Ondas Gravitacionales e Investigación Asistida por IA

The prediction, and the *gap*.

From a census of local supermassive black holes you can predict the nanohertz gravitational-wave background — with **no free parameters**. The prediction comes out **a factor of a few low** in black-hole mass density, and no merger history can close the gap.

- how the clocks work, and why nanohertz;
- the Hellings–Downs curve, derived;
- how the black holes are actually counted — the M – σ relation;
- the prediction against the measurement;
- three ways out, and how the evidence has already narrowed them.

Every step is checkable, and nothing in the chain is fitted to the PTA measurement.

The background of the slide is a dark, deep blue space filled with numerous small, white, four-pointed stars. Overlaid on this is a faint, light blue grid of lines that curve and warp around two central, swirling galaxy-like structures. These structures are composed of concentric, swirling bands of color, including shades of orange, yellow, and purple, suggesting intense gravitational fields or the formation of black holes. The overall effect is one of cosmic scale and dynamic movement.

SECTION I

The nanohertz sky

Millisecond pulsars as *clocks*.

A millisecond pulsar spins hundreds of times a second, and its pulse arrival times can be predicted to hundreds of nanoseconds over decades. You fit a **timing model** — spin, spin-down, position, proper motion, binary parameters — and subtract it. What is left is the **residual**, and the residual is the observable.

A gravitational wave perturbs the light travel time. The fractional frequency shift is

$$z(\hat{p}) = \frac{1}{2} \frac{\hat{p}^i \hat{p}^j h_{ij}}{1 + \hat{\Omega} \cdot \hat{p}} \bigg|_{\text{Earth}} - \frac{1}{2} \frac{\hat{p}^i \hat{p}^j h_{ij}}{1 + \hat{\Omega} \cdot \hat{p}} \bigg|_{\text{pulsar}}$$

The integral along the pulse's path leaves the metric at its **two endpoints**: the **Earth term** — h_{ij} here, now — and the **pulsar term** — h_{ij} at the pulsar, one light-travel-time ago. $\hat{\Omega}$ is the direction the wave propagates, $\hat{p}$ the direction **to the pulsar**.

One measurement, two *wavelengths*.

A PTA and LIGO are the **same measurement**: a wave perturbing a light travel time between two free masses.

LIGO

The wavelength is **long** compared with the 4 km arm: the whole apparatus sits in one phase of the wave, and the response is the strain itself.

A PULSAR PAIR

The "arm" is kiloparsecs and the wavelength is **short** compared with it: the two endpoints feel uncorrelated phases, which is exactly the Earth-term/pulsar-term split.

Same formula, one dimensionless ratio — wavelength over separation — deciding which limit you are in.

Why *nanohertz*.

A dataset of length T sampled every Δt measures

$$f_{\min} \sim \frac{1}{T}, \quad f_{\max} \sim \frac{1}{2\Delta t}$$

Fifteen years gives $f_{\min} \approx 2$ nHz; fortnightly cadence gives $f_{\max} \approx 400$ nHz. The band is set by the experiment, not the physics — which is why a PTA is a decades-long project.

At those frequencies a circular binary of $\sim 10^9 M_{\odot}$ is about a year of orbits from merger, and the sources are **supermassive black-hole binaries**.

What one source *looks like*.

The day-1 amplitude formula, at these masses and distances:

$$h_0 = \frac{2 (G\mathcal{M}_c)^{5/3} \Omega^{2/3}}{c^4 d} \approx 1.3 \times 10^{-15} \quad \text{for } \mathcal{M}_c = 10^9 M_\odot \text{ at } 150 \text{ Mpc}$$

That is the scale everything today sits at — six orders of magnitude louder than a LIGO event in strain, and still at the edge of what fifteen years of timing can see, because you wait years per cycle.



SECTION II

Hellings and Downs

Hellings and Downs, *derived*.

Put one source at the pole. The $\sin^2 \theta$ of the antenna pattern **cancels** the $(1 - \cos \theta)$ of the propagation denominator:

$$\tilde{z} = \frac{1}{2}(1 + \cos \theta)(\cos 2\phi \tilde{h}_+ - \sin 2\phi \tilde{h}_\times)$$

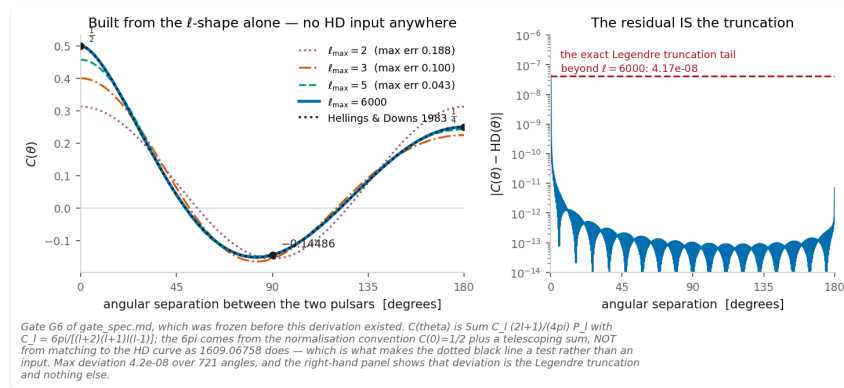
The ϕ integral kills every m except ± 2 — the spin-2 signature. The θ integral is

$$\int_{-1}^1 (1 + \mu) P_\ell^2(\mu) d\mu = 4$$

independent of ℓ . Hence

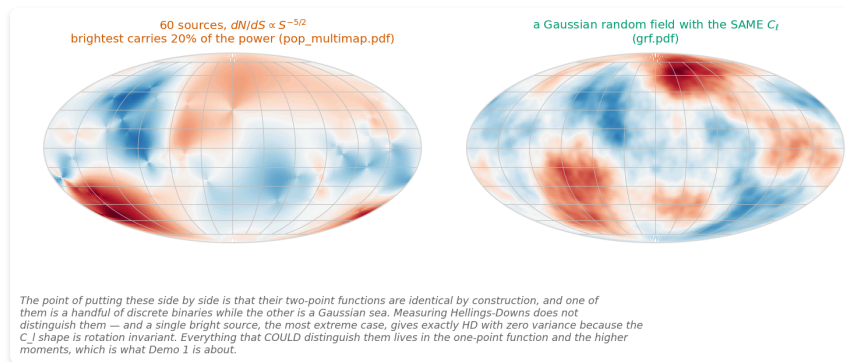
$$C_\ell^z = \frac{6\pi}{(\ell + 2)(\ell + 1)\ell(\ell - 1)}$$

and the curve everybody plots is that same set of numbers, summed back up:



Built from the multipole shape alone. The 1983 curve is a test, not an input.

Many sources, same *curve*.



A single bright source and a Gaussian random field: identical normalised correlation.

That was **one** source. The background is millions.
Two facts finish the derivation:

- each source arrives with a **random phase**, so in the *power* they add **incoherently** — no cross terms survive the average;
- C_ℓ is **rotationally invariant**, so sources at different sky positions simply **add their C_ℓ** — the sum has the same ℓ -shape as one source.

One source's spectrum *is* the background's. Verified here across six random source directions: agreement to 7×10^{-6} .

The same fact read backwards: measuring the Hellings-Downs correlation tells you the signal is a **spin-2 metric perturbation**. It does not tell you there are many sources.

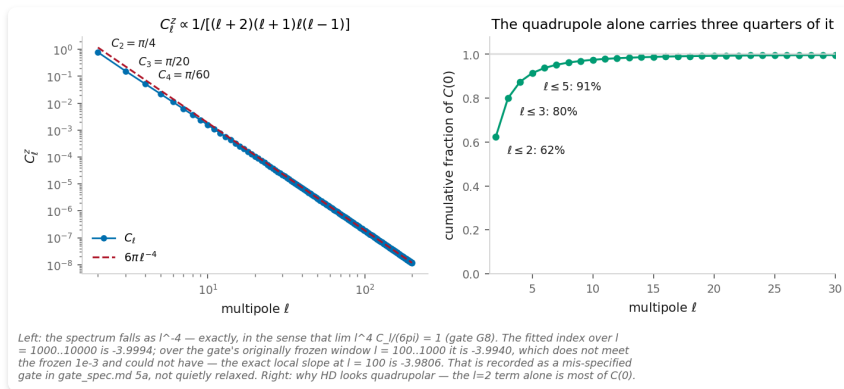
The Earth term is what *correlates*.

The redshift has two endpoint terms. What we just computed is the correlation of the **Earth terms** of two *distinct* pulsars — the same h_{ij} , here and now, projected along two different directions.

The **pulsar term** cannot join: each pulsar's distance is unknown to within many wavelengths, so its phase is effectively **random per pulsar**, and it averages out of any cross-correlation.

Where it does survive is the **self**-correlation: there the pulsar term adds its own power, doubling $C(0)$ from $\frac{1}{2}$ (two distinct pulsars at zero separation) to 1 (one pulsar with itself) — a diagonal contribution, and nothing else.

A very red spectrum.



The multipoles of the redshift map, against the exact fractions.

C_l falls as l^{-4} — exactly, in the sense that $\lim_{l \rightarrow \infty} l^4 C_l / 6\pi = 1$.

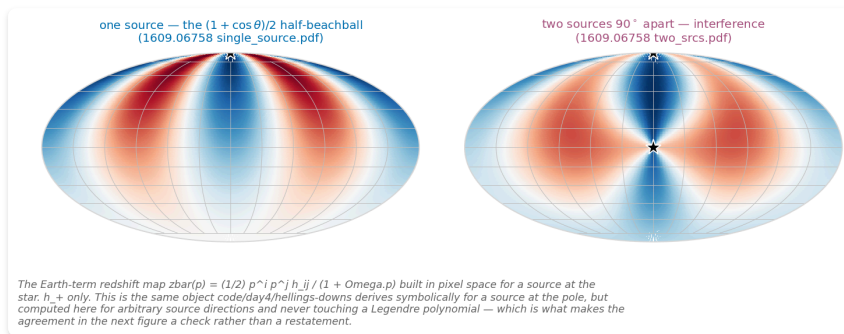
So Hellings–Downs is a **handful of multipoles**. The quadrupole alone carries most of $C(0)$, and by $\ell_{\max} = 5$ the curve is right to a few per cent.

Exact values:

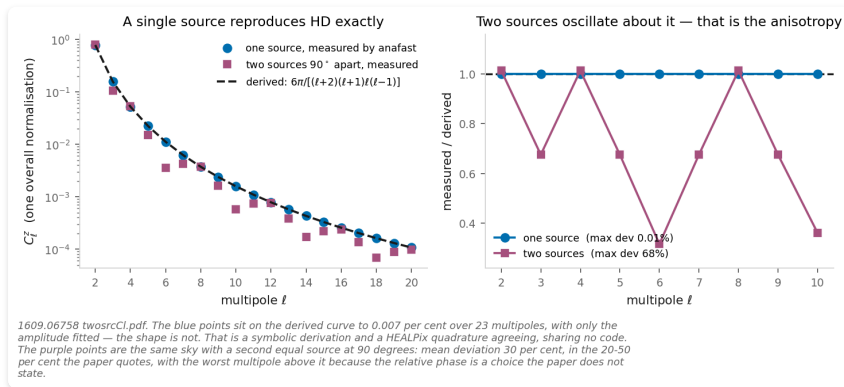
$$C_2 = \frac{\pi}{4}, \quad C_3 = \frac{\pi}{20}, \quad C_4 = \frac{\pi}{60}$$

and the monopole and dipole are **absent**, not subtracted.

Skies, and their *spectra*.



Source skies in pixel space.



Their measured multipoles against the derived spectrum.

The same physics, computed a second way: build the map in pixel space, transform it, measure C_ℓ , and compare with the spectrum derived symbolically.

They agree to 7×10^{-5} over 23 multipoles, with **only the amplitude** fitted — the shape is untouched. The two computations share no code: one integrates Legendre functions, the other transforms a pixelised sky.

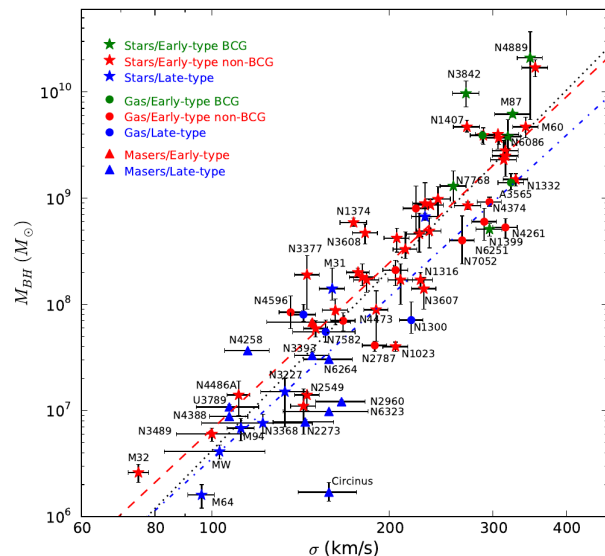
Add a second source and the measured spectrum **oscillates** about the smooth curve by tens of per cent. That oscillation *is* the anisotropy — hold that thought for the sky-power slide.



SECTION III

Counting the black holes

The M– σ relation.



McConnell & Ma 2013, ApJ 764, 184, Fig. 1

A black-hole mass is measured **dynamically**: resolve stars or gas inside the sphere of influence,

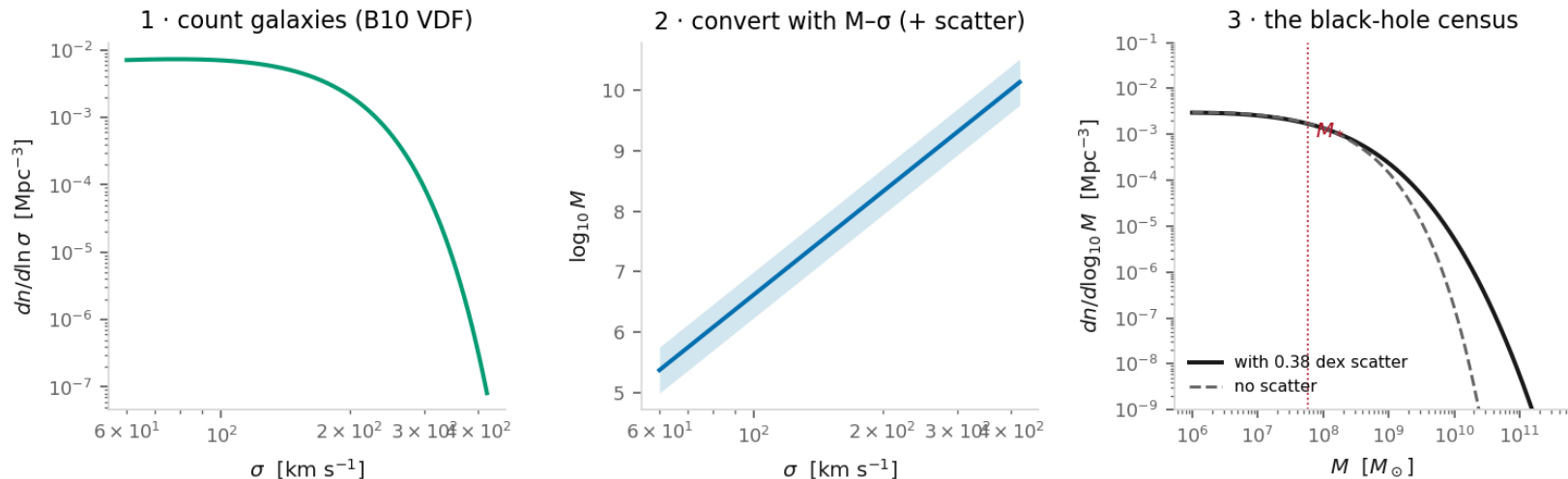
$$r_{\text{infl}} = \frac{GM}{\sigma^2} \approx 11 \text{ pc} \quad (10^8 M_{\odot}, \sigma = 200 \text{ km s}^{-1})$$

— an arcsecond at a couple of Mpc, milliarcseconds further out. So direct masses exist only **nearby**: seventy-two galaxies, each one named on that plot. They correlate tightly with the host's velocity dispersion:

$$\log_{10} M = a + b \log_{10} \frac{\sigma}{200 \text{ km s}^{-1}}$$

with $a = 8.32$, $b = 5.64$, and an **intrinsic scatter** of 0.38 dex — a dex being a factor of ten, so the scatter is a factor ~ 2.4 either way. That fit is the black dotted line, and the spread around it is what the rest of the day inherits.

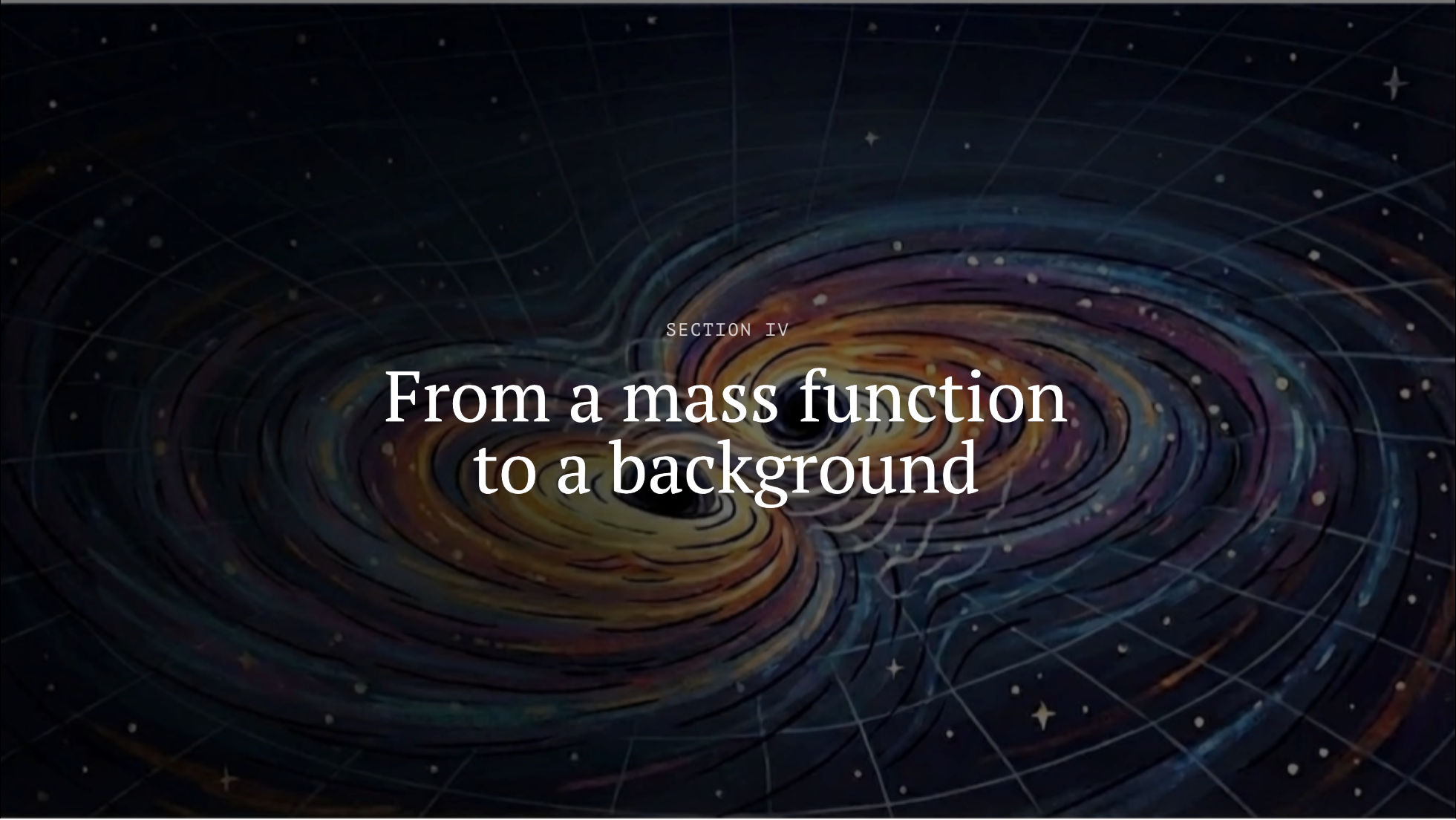
How the census is *assembled*.



The whole census in three steps: a galaxy count in σ , the scaling relation with its scatter, and the resulting mass function. The scatter matters most exactly where the GW background integral lives — the heavy tail.

Count galaxies in σ , convert through M - σ with its scatter, read off the black-hole mass function.

Nobody counts black holes directly. You count **galaxies** — the velocity-dispersion function — and push it through the relation. The knee lands at $M_* = 5.85 \times 10^7 M_\odot$, and the **scatter matters most in the heavy tail**, exactly where the gravitational-wave integral will live.

The background of the slide is a dark, deep blue space filled with numerous small, white stars of varying sizes. Overlaid on this starry field are several large, concentric, swirling patterns that resemble gravitational wells or the structure of spiral galaxies. These patterns are rendered in shades of blue, purple, and orange, with the most intense colors concentrated in the centers of the swirls. A faint, white grid of lines is visible across the entire image, suggesting a coordinate system or a map of the universe.

SECTION IV

From a mass function to a background

Phinney's theorem.

The energy density per logarithmic frequency equals the comoving density of event **remnants** times the redshifted energy each event radiated:

$$\rho_c c^2 \Omega_{\text{gw}}(f) = \int_0^\infty N(z) \frac{1}{1+z} \left(f_r \frac{dE_{\text{gw}}}{df_r} \right) \bigg|_{f_r=f(1+z)} dz$$

The identification that makes it powerful: the **remnant** mass function is the merger mass function integrated over z and q . So the prediction does not depend on the merger history — and it is an **upper limit** if black holes merged more than once.

Phinney's own remark: "*to derive this one need know nothing about gravitational radiation except that it has twice the orbital frequency, and that the orbital binding energy is removed by the gravitational radiation.*" Kepler plus binding energy indeed gives $dE/d \ln f_r = \frac{1}{3G} (G\mathcal{M})^{5/3} (\pi f_r)^{2/3}$, checked symbolically in the day's reproduction.

$h_c \propto f^{-2/3}$ is kinematic.

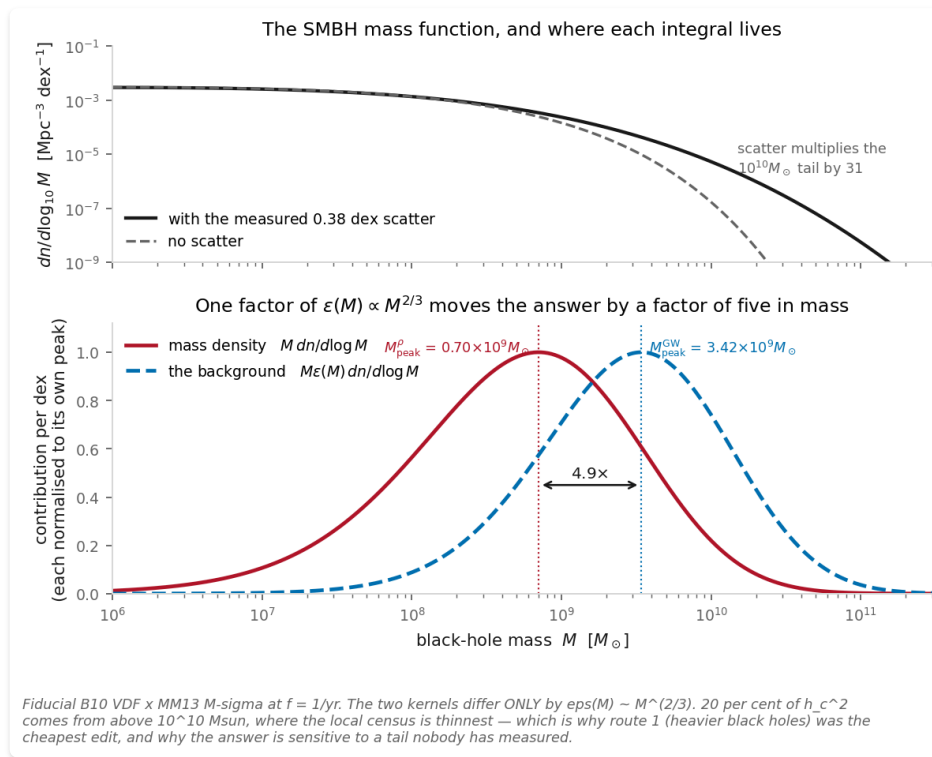
For an energy spectrum $dE/d\ln f \propto f^p$,

$$h_c \propto f^{(p-2)/2}$$

for **any** mass function — the whole population drops out of the exponent. Circular GW-driven inspiral has $p = 2/3$, hence $-2/3$.

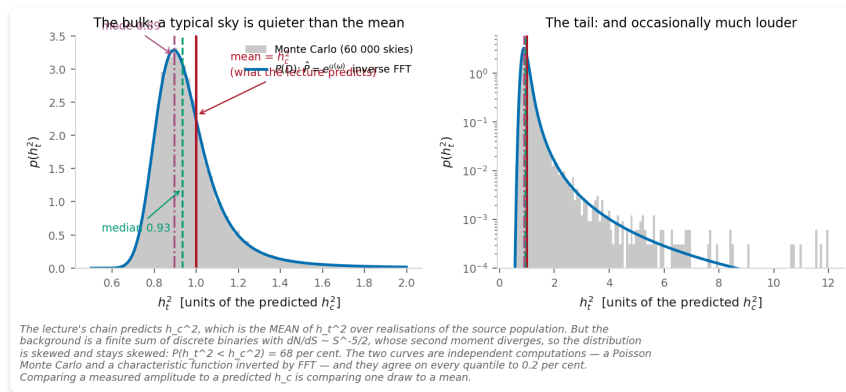
So a measured departure from $-2/3$ is evidence about **environments or eccentricity**. It says nothing about the population's mass — the most common misreading of the PTA spectral index.

Which black holes *carry it*?



The mass-density and background kernels, and the gap between their peaks.

The typical sky sits *below the mean*.



The distribution of the squared strain over source realisations, two independent constructions.

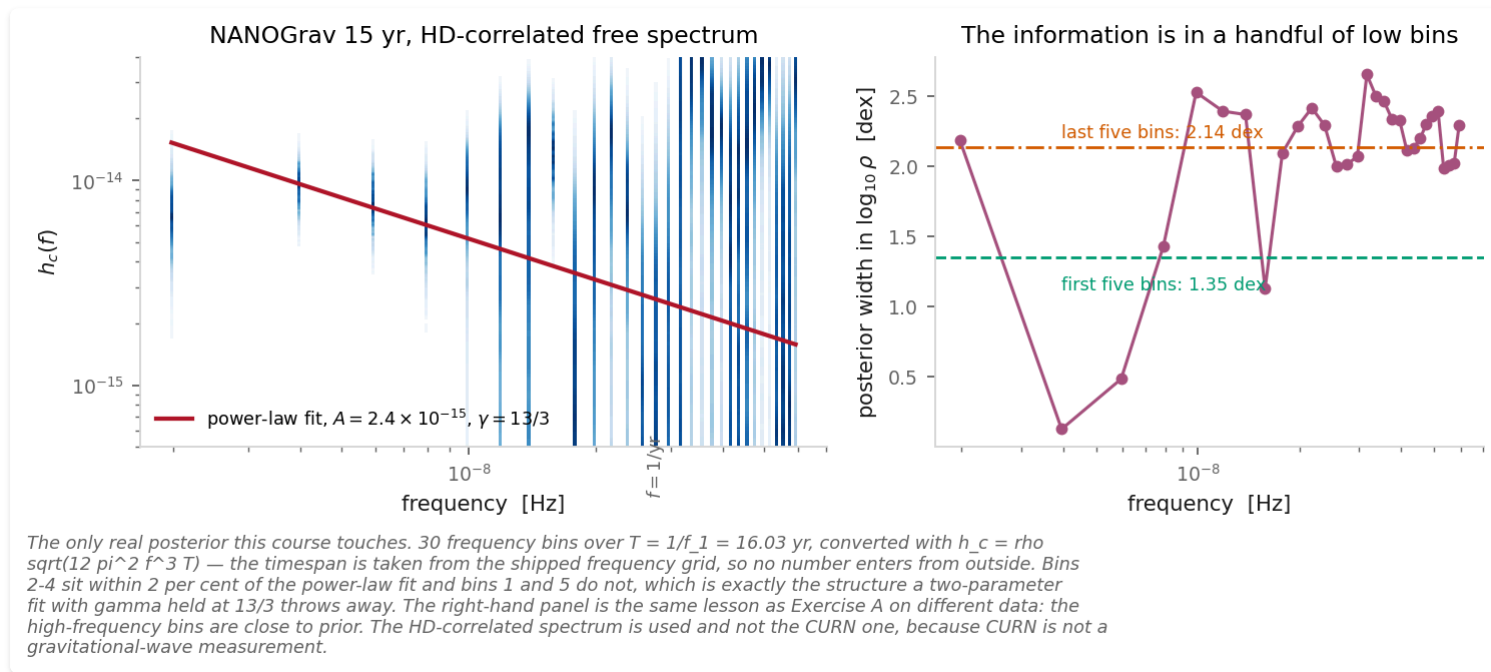
h_c^2 is the **mean** of h_t^2 over realisations of the source population. A PTA measures **one draw** — and when rare heavy sources dominate, the draw is usually below the mean:

$$\text{mode } 0.89 h_c^2, \quad \text{median } 0.93, \quad P(h_t^2 < h_c^2) = 68\%$$

With $dN/dS \propto S^{-5/2}$ the second moment diverges, so no amount of averaging Gaussianises it — and the shortfall can be **large enough to rule out** scenarios dominated by the very heaviest sources.

This is Sato-Polito & Zaldarriaga's point, and it returns in the ending: the free spectrum's fluctuations carry information the mean does not.

And the real *thing*.



NANOGrav 15 yr, HD-correlated free spectrum — the per-bin amplitudes fitted with no spectral model assumed. One posterior per frequency bin, of the kind day 3 built, and the only one in this course that was measured rather than simulated.

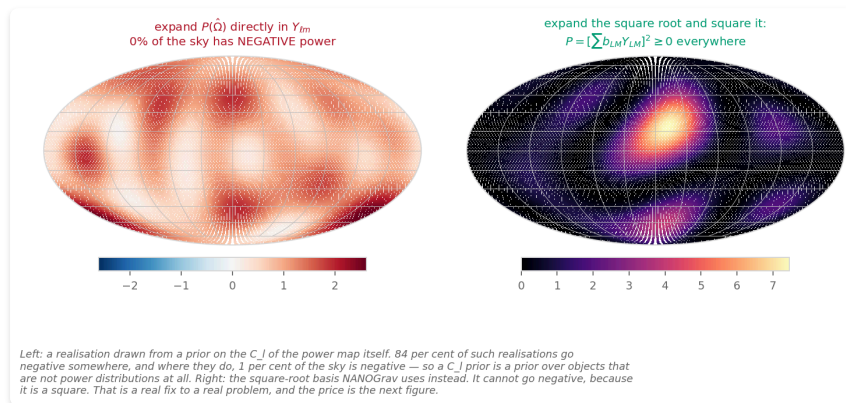
What the measurement *establishes*.

EVIDENCE, NOT DETECTION — all major PTA collaborations independently report the Hellings–Downs correlation at the $3\text{--}4\sigma$ level. Strong, consistent — and not yet the five-sigma standard the field calls a detection.

THE SPECTRAL INDEX — the fits are broadly consistent with the kinematic $\nu^{-2/3}$ (in timing-residual power, $\gamma = 13/3$), but the free spectrum carries structure a two-parameter power law does not represent.

RED NOISE, AND CONFUSION — each pulsar also carries **intrinsic red noise** with the same steep character as the signal. In a single pulsar the two are indistinguishable; only the **cross-pulsar correlation** — the curve we derived — separates a background from a sky of noisy clocks. That is why the correlation, not the spectrum, is the discovery statistic.

The power of the sky.



Source skies are positive; a Gaussian expansion of the power is not.

The anisotropy observable is the **source-power sky** and its spectrum:

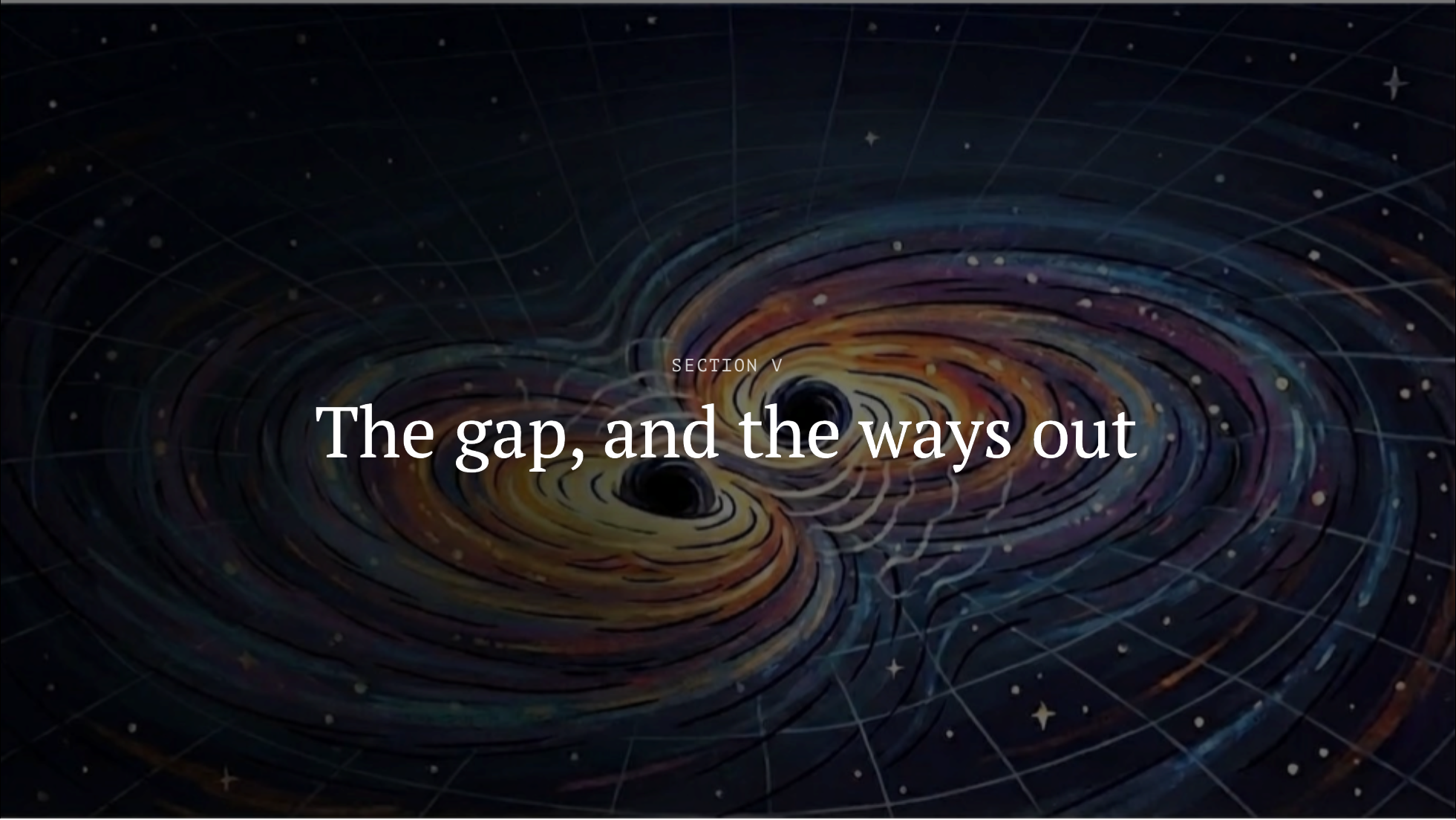
$$P(\hat{\Omega}) = \sum_a q_a \delta^{(2)}(\hat{\Omega}, \hat{n}_a), \quad C_\ell^{(P)}$$

For isotropically placed sources,

$$\mathbb{E} \left[\frac{C_\ell^{(P)}}{C_0^{(P)}} \right] = \frac{1}{N_{\text{eff}}} \quad \text{independent of } \ell$$

— one dominant source gives a **flat** spectrum, and $P(\hat{\Omega}) \geq 0$ constrains any expansion you might try.

This $C_\ell^{(P)}$ is not the derivation's C_ℓ^z — one is the power on the sky, the other the coherent redshift map. This afternoon's exercise builds such skies and measures what a published anisotropy limit is actually a statement about.



SECTION V

The gap, and the ways out

The gap.

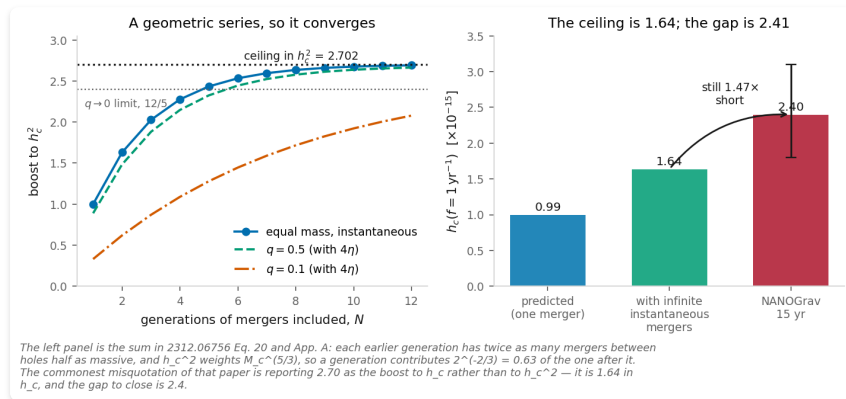
	value
M_*	$5.85 \times 10^7 M_\odot$
$\rho_\bullet$	$4.54 \times 10^5 M_\odot \text{ Mpc}^{-3}$
predicted $h_c(1/\text{yr})$	1.0×10^{-15}
measured	$2.4^{+0.7}_{-0.6} \times 10^{-15}$
the gap	$2.4 \times$ in strain, $5.8 \times$ in mass density

Nothing in the prediction is fitted to the measurement.

Both rows are quoted at 1/yr because that is **how the measurement gets reported**, not because it is the frequency anybody measures. The band a timing array is actually sensitive to sits below it, and getting to 1/yr is a power-law extrapolation.

The mass density is independently consistent with the Soltan argument — the accreted mass counted from the quasar luminosity function — which is what makes this interesting rather than a calibration problem.

The merger ceiling.



The geometric series of generations, and where every relaxation moves it.

Could past mergers hide the mass? Go back one generation in an equal-mass tree: twice as many events, between holes half as massive. h_c^2 weights $\mathcal{M}_c^{5/3}$, so a generation contributes $2 \times 2^{-5/3} = 2^{-2/3}$ of the next. Geometric, hence convergent:

$$\sum_{N=1}^{\infty} 2^{-2(N-1)/3} = \frac{1}{1 - 2^{-2/3}} = 2.702 \Rightarrow h_c \text{ ceiling} =$$

The commonest misquotation reports 2.70 as the boost to h_c rather than to h_c^2 .

Unequal masses give less ($12/5$ as $q \rightarrow 0$); accretion between mergers gives less still ($\kappa = \frac{1}{2} \Rightarrow 1.25$). **Every relaxation moves the ceiling down** — and the ceiling is 1.64 against a gap of 2.4.

1 — Heavier black holes than the census *shows*.

Proposed first, and proposed **because it is the cheapest edit**: the high-mass tail is where there are fewest observations, so a modest change there does the most work.

It makes a falsifiable prediction: $M_{\text{peak}} \sim 3 \times 10^{10} M_{\odot}$ means about **60 black holes above** $10^{10} M_{\odot}$, and a couple above $10^{11} M_{\odot}$, within 100 Mpc — which **MASSIVE**, the dedicated survey of the most massive nearby galaxies, can go and count.

Now less likely. Heavy means few, few means large Poisson fluctuations in the spectrum, and the NANOGrav free spectrum disfavors that: $M_{\text{peak}} \lesssim 10^{10} M_{\odot}$ at 2σ .

Same data, one extra piece of information, opposite conclusion — a hypothesis proposed on grounds of economy, then narrowed by evidence. That is how inference is supposed to work.

2 — A census more *uncertain*.

The prediction is exquisitely sensitive to a **relative** systematic offset between the small black-hole catalogue and the large galaxy catalogue:

$$\frac{h_{c,\delta}}{h_c} = 10^{\frac{5}{6} b_{\bullet} \delta}$$

Every symbol is one you have met: $b_{\bullet}$ is the **M- σ slope** ($b = 5.64$, two sections ago), and δ is a relative offset in $\log_{10} \sigma$ between the two catalogues, in dex. An offset of only **0.03 dex** moves h_c by $\times 1.38$ — the scale of the NANOGrav 90% interval.

And the one correction that *is* well understood — aperture, σ at R_e versus $R_e/8$ — goes the **wrong way**, reducing h_c by $\times 0.68$.

3 — The measurement is a bit *off*.

A live possibility, stated as one.

The evidence at fifteen years is *evidence*, not a detection, and the free spectrum carries real structure that a two-parameter power-law fit does not represent.

To take away.

- **A pulsar pair is an interferometer with a kiloparsec arm.** Same measurement as LIGO; the wavelength-to-arm ratio decides the limit.
- **Hellings–Downs is geometry.** A handful of multipoles, $C_\ell \propto \ell^{-4}$, derived — and it certifies spin 2, not a source count. What correlates between pulsars is the Earth term.
- **The census is a galaxy count pushed through $M-\sigma$.** The knee is $M_* \approx 6 \times 10^7 M_\odot$, and the heavy tail — where the background lives — is where the scatter and the systematics bite.
- **The parameter-free prediction falls short of the measurement** by about 2.4 in strain, and the merger ceiling of 1.64 means no merger history closes it.
- **Three ways out, already narrowed:** heavier holes (disfavoured by the spectrum's own fluctuations), a shiftier census (a few hundredths of a dex is enough), or a measurement that settles as it lengthens.